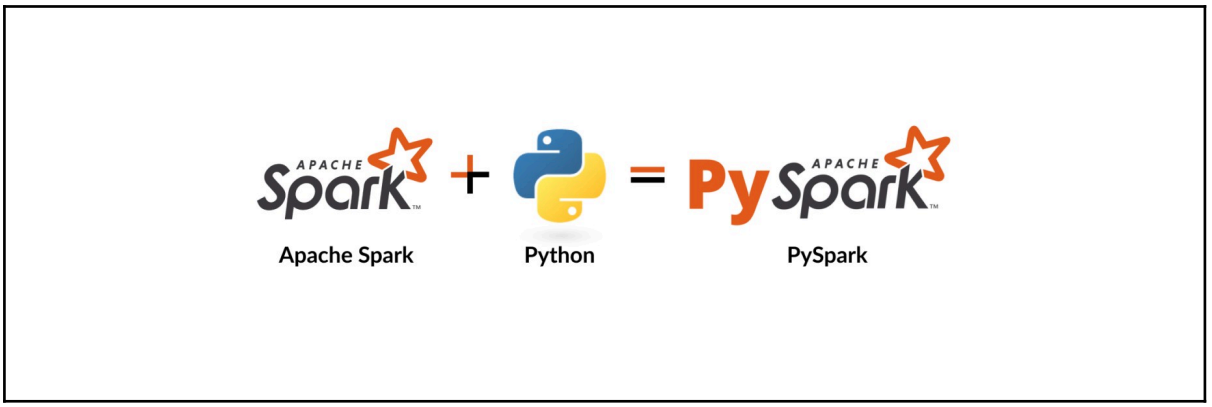
Introduction to Apache Spark and Pyspark

PySpark is based on Apache Spark, which is an open source analytics engine for large scale data processing.



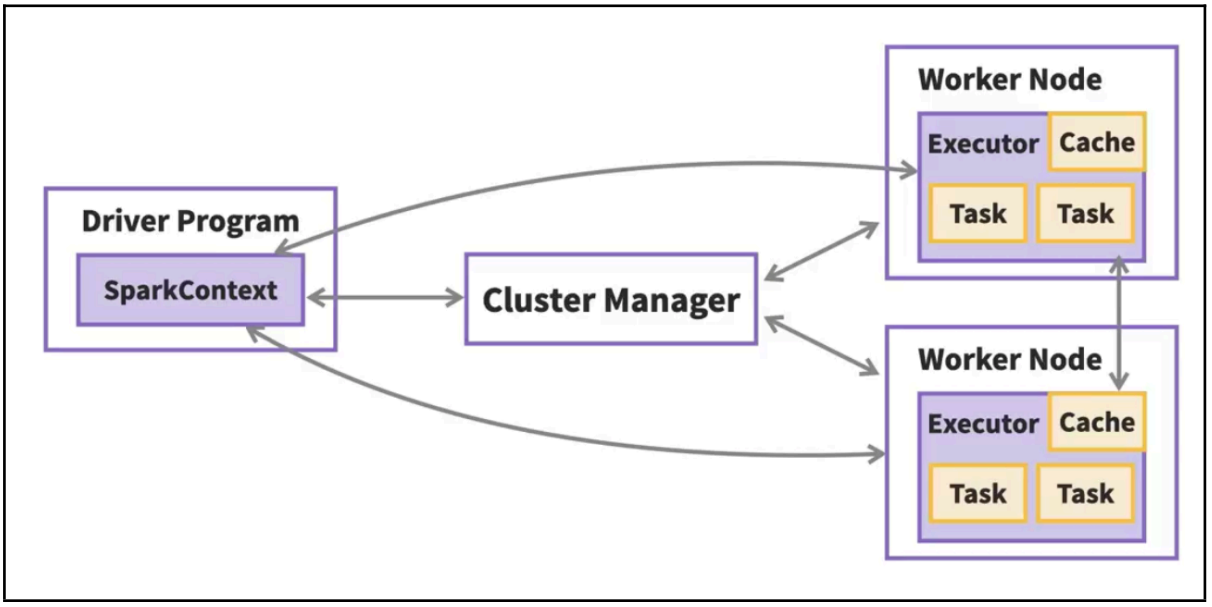
Apache Spark was developed around 2009 and officially became an open source project of the apache software foundation in 2013. Spark is optimized for parallel and highly fault tolerant processing of massive datasets on a cluster of machines.

What is a Computer Cluster?

A computer cluster is a group of interconnected machines that work together as one system. Each machine is called a node and their coordination is handled by a cluster manager. This setup allows large computations to be split across nodes, speeding up processing compared to a single machine.

How does a spark cluster work?

The below diagram shows how spark runs on a cluster. The driver program coordinates the work and talks to the cluster manager, which assigns tasks to worker nodes. Each worker runs an executor inside a Java virtual machine (JVM), allowing spark to process massive datasets in parallel.



Do you need a cluster to use spark?

In short, no. Spark also works on a single machine, which means you can simply use your own computer for development and testing but it's recommended to use computer clusters for production.

Apache Spark Vs. Hadoop MapReduce

Apache Spark	Hadoop MapReduce
Does not have its own native file system.	Has a native file system called Hadoop Distributed File System (HDFS).
More suitable for real-time processing.	Better suited for batch jobs.
Lower latency due to in-memory data storage during processing.	Higher latency due to data writes at each processing step.
Built-in ML library.	No native ML capabilities.